



5.4 Kinematics using calculus

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

For motion in a straight line, displacement $s(t)$, velocity $v(t)=ds/dt$, and acceleration $a(t)=dv/dt=d^2s/dt^2$. Conversely, velocity is the integral of acceleration, and displacement is the integral of velocity: $v=\int a \, dt$, $s=\int v \, dt$ (with constants determined by initial conditions). The object is momentarily at rest when $v=0$, and distance travelled (as opposed to displacement) requires accounting for direction changes using $|v|$.

Worked example: follow the reasoning

A particle has velocity $v(t)=3t^2-12t+9$ (m/s). Find its acceleration at $t=1$ and the times when it is at rest.

$a(t)=dv/dt=6t-12$. At $t=1$: $a(1)=6-12=-6$ m/s².

At rest: $3t^2-12t+9=0 \rightarrow t^2-4t+3=0 \rightarrow (t-1)(t-3)=0 \rightarrow t=1$ or $t=3$.

Quick check · multiple choice

A particle has displacement $s(t)=t^3-3t$. What is its velocity at $t=2$?

- A 3
- B 6
- C 9
- D 12

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A particle moves in a straight line so that its velocity at time t seconds ($t \geq 0$) is [8]
 $v(t) = t^2 - 6t + 8$ (m/s).
(a) Find the times when the particle is at rest.
(b) Find the acceleration when $t=4$.
(c) Given that the particle starts ($t=0$) at displacement $s=0$, find an expression for $s(t)$, and find the displacement at $t=3$.

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $t^2 - 6t + 8 = 0 \rightarrow (t-2)(t-4) = 0 \rightarrow t=2$ or $t=4$.

(b) $a(t) = dv/dt = 2t - 6$. At $t=4$: $2(4) - 6 = 2 \text{ m/s}^2$.

(c) $s(t) = \int (t^2 - 6t + 8) dt = t^3/3 - 3t^2 + 8t + C$. Since $s(0)=0$, $C=0$. $s(t) = t^3/3 - 3t^2 + 8t$. At $t=3$:
 $27/3 - 3(9) + 8(3) = 9 - 27 + 24 = 6 \text{ m}$.

Quick check answer

C. $v(t) = ds/dt = 3t^2 - 3$. At $t=2$: $3(4) - 3 = 9$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.